Applied ML

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Text Preprocessing

- tabular data is structured data having rows/columns

- text, image, audio, are unstructured data.

Text - sentiment analysis, tweets classification, news classification

* Sentiment analysis

"I like this phone, it has very good camera" → [model] → +ve ✓ / -ve / neutral

Text	target
1. The phone is good	+ve
2. Battery is bad	-ve
3. size is great	+ve
4. camera is high quality	+ve
5. I used it for 1 year	neutral
6. Charging is very slow	-ve

} Corpus

ML models only accept input in numerical format.

We need to convert this text data into numerical format before using it for ML training.

Text data $\xrightarrow{\text{text preprocessing}}$ numerical data

↓

ML training

Tokenization

- Breaking text / documents into words / tokens.
- convert text into lower case before tokenization

eg. I like the food ⟶ [i, like, the, food]

There was a big tiger → [there, was, a, big, tiger]

Create vocabulary for given corpus

- vocab is set of all unique tokens in given corpus.

Text	target	tokenized format
1) It was a good day	+ve	[it, was, a, good, day]
2) service was bad	-ve	[service, was, bad]
3) food is great	+ve	[food, is, great]
4) taste was good	+ve	[taste, was, good]
5) parking was not clean	-ve	[parking, was, not, clean]

vocab: {it, was, a, good, day, service, bad, food, is, great, taste parking, not, clean}

Text cleaning - Stop words removal

- Stop words are the words which don't contribute to the sentiment / intent of the text.

eg. was, there, is, a, an, the, that, it, etc

- Hence, these type of words are removed from vocabulary.

- This reduces the vocabulary size which reduces complexity of the task.

eg. it was a good day +ve

vocab after stop words removal -
{ good, day, service, bad, food, great, taste, parking, not, clean }

natural language tool kit

It was a great idea ⟶ nltk ⟶ [great, idea]

tokenization
+
stop words removal

Text	tokenized + stop words removal
it was a good day	[good, day]
service was bad	[service, bad]
food is great	[food, great]
taste was good	[taste, good]
parking was not clean	[parking, not, clean]

Create Document Term Frequency Table

vocab = { good, day, service, bad, food, great, taste, parking, not, clean }

doc	good	day	service	bad	food	great	taste	parking	not	clean
1	1	1	0	0	0	0	0	0	0	0
2	0	0	1	1	0	0	0	0	0	0
3	0	0	0	0	1	1	0	0	0	0
4	1	0	0	0	0	0	1	0	0	0
5	0	0	0	0	0	0	0	1	1	1

doc : taste was good

this is numerical representation of given document

taste was good

↓

[taste, good]

↓

$$\underbrace{\begin{matrix} [1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0] \\ x_1 & x_2 & x_3 & x_4 & x_5 & x_6 & x_7 & x_8 & x_9 & x_{10} \end{matrix}}_{\text{10 input features}} \longrightarrow \boxed{\text{model}} \longrightarrow \text{target}$$

parking was not clean	-ve	corpus
you can park there	neutral	
I parked in front.	neutral	

vocab = { parking, not, clean, park, parked, front }

when the word is present in multiple forms in the vocab, then only include its base word in vocab.

Lemmatization

- converting any given word into its base form

eg.

parking → park	education → educate
parked → park	teacher → teach
park → park	service → serve

vocab = { parking, not, clean, park, parked, front }

↓ after lemmatization

vocab = { park, not, clean, front }

text : parking is clean in front

↓ tokenization

[parking, is, clean, in, front]

↓ stop word removal

[parking , clean, front]

↓ lemmatization

[park, clean, front]

text: [park, clean, front]

vocab = {park, not, clean, front}

	park	not	clean	front
[park, clean, front] =	[1	0	1	1]

lemmatization - done using nltk library

- Lemmatization is a slow process and can become a bottleneck in case of large corpus.

- in such cases we do stemming instead of lemmatization.

Stemming

- removes suffixes from words to make them shorter and similar to base words.

eg.

parking → park
coming → com ≈ come

study → stud
studying → stud
studies → stud
studied → stud

} different forms of word get converted same form

coming $\xrightarrow{\text{lemma}}$ come

$\xrightarrow{\text{stem}}$ com

student $\xrightarrow{\text{lemma}}$ study

$\xrightarrow{\text{stem}}$ stud

organisation $\xrightarrow{\text{lemma}}$ organise

$\xrightarrow{\text{stem}}$ organis

lemmatization – get a meaningful word as output

eg. coming → come

Stemming – might not get meaningful word as output

eg. coming → com

☆ Stemming is very fast and suitable for large corpus.